Kubernetes Assignment

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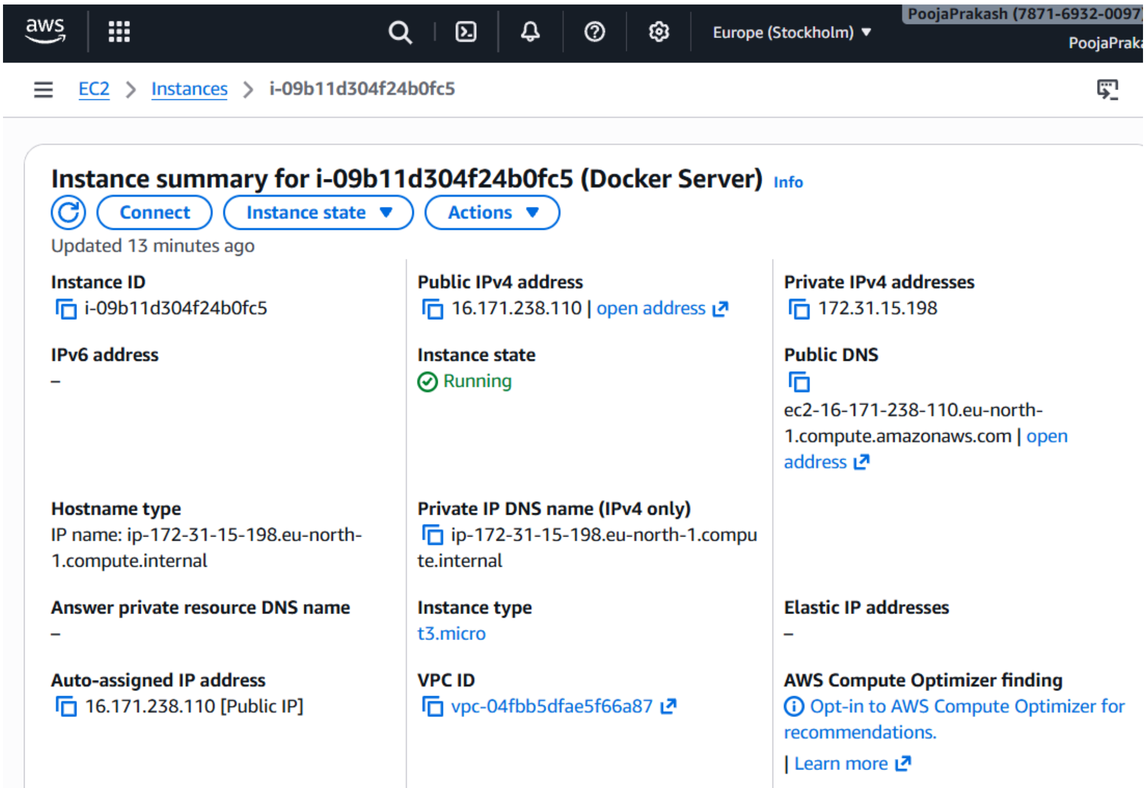
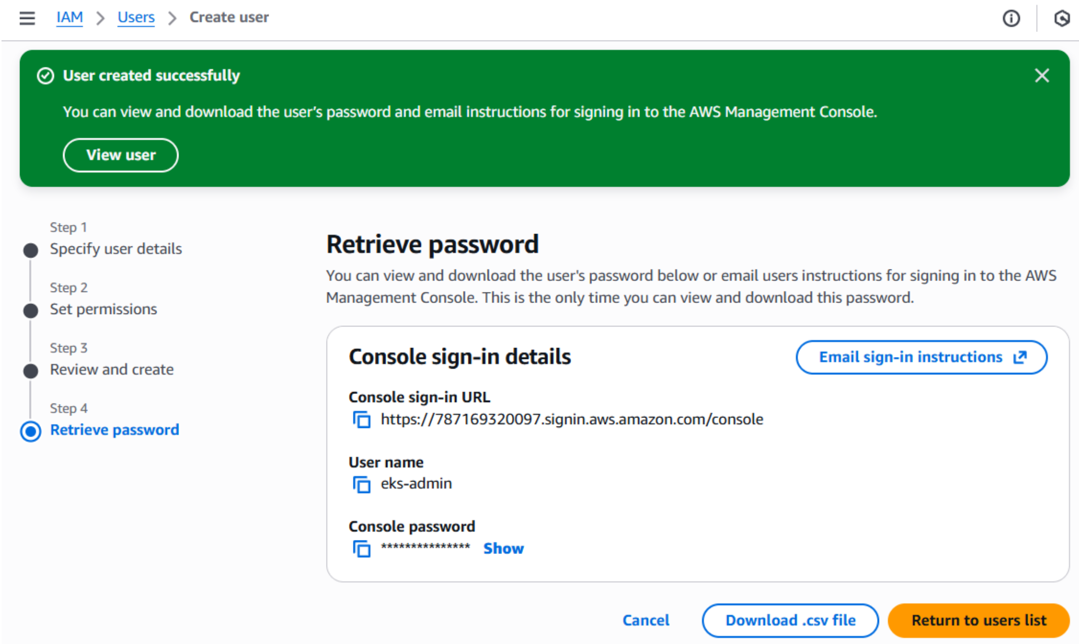
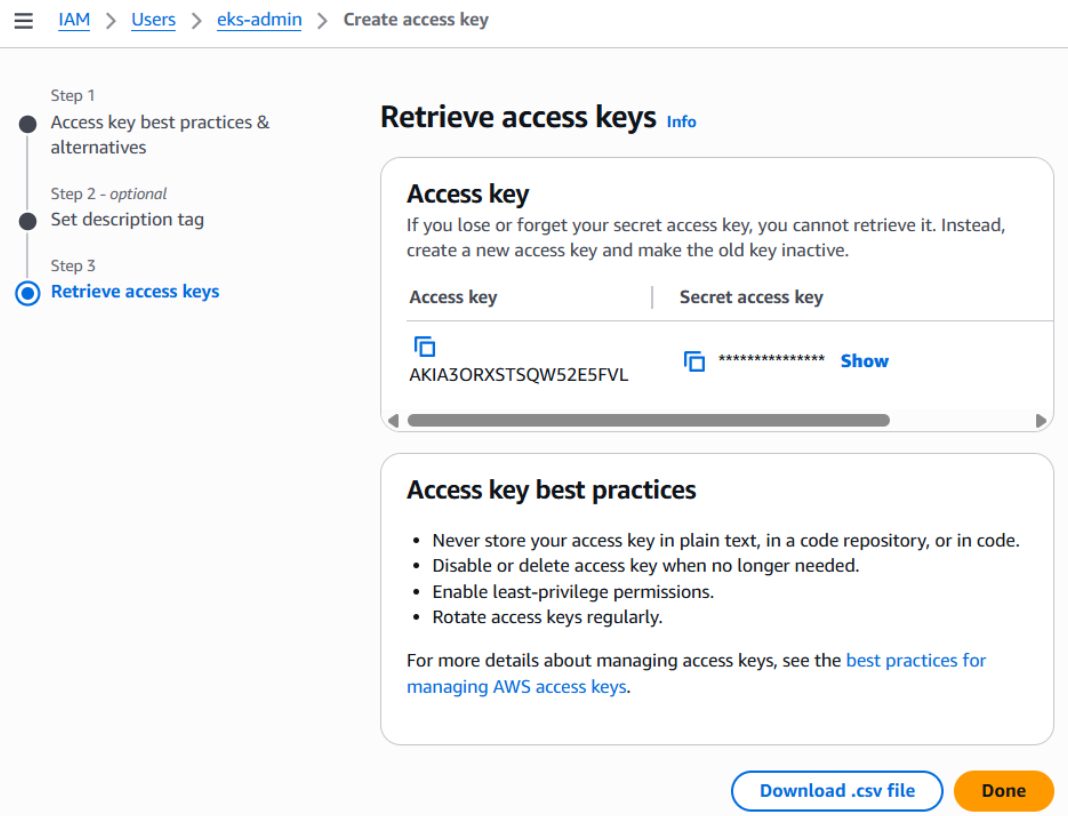
1. L1 - Create Deployment Controller Object to Deploy the Application Image Created in Docker Module and Expose it to the Internet.

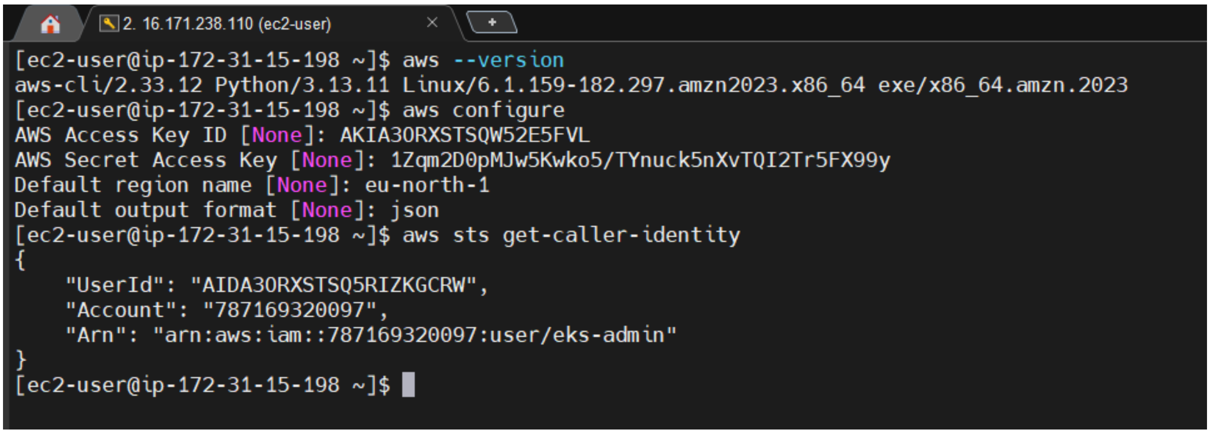


- Create an IAM user to communicate with the Kubernetes cluster.



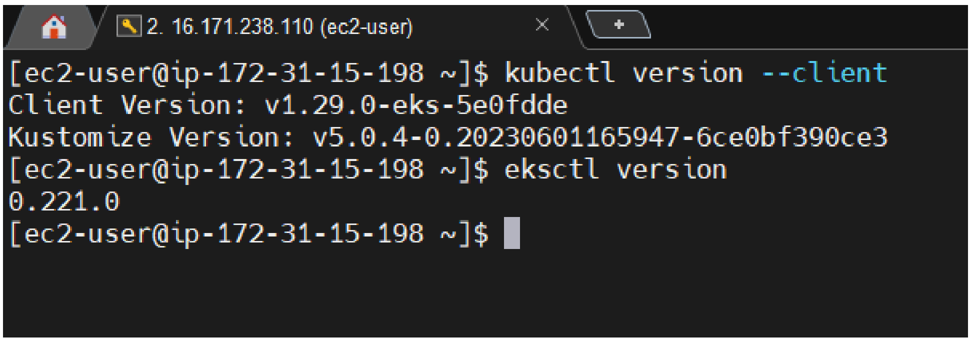


- Install the AWS CLI and configure it with the required credentials.




```
[ec2-user@ip-172-31-15-198 ~]$ aws --version
aws-cli/2.33.12 Python/3.13.11 Linux/6.1.159-182.297.amzn2023.x86_64 exe/x86_64.amzn.2023
[ec2-user@ip-172-31-15-198 ~]$ aws configure
AWS Access Key ID [None]: AKIA3ORXSTSQW52E5FVL
AWS Secret Access Key [None]: 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
Default region name [None]: eu-north-1
Default output format [None]: json
[ec2-user@ip-172-31-15-198 ~]$ aws sts get-caller-identity
{
    "UserId": "AIDA3ORXSTSQ5RIZKGCRW",
    "Account": "787169320097",
    "Arn": "arn:aws:iam::787169320097:user/eks-admin"
}
[ec2-user@ip-172-31-15-198 ~]$
```

- Install `kubectl` and `eksctl` to manage Kubernetes clusters on AWS.




```
[ec2-user@ip-172-31-15-198 ~]$ kubectl version --client
Client Version: v1.29.0-eks-5e0fdde
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
[ec2-user@ip-172-31-15-198 ~]$ eksctl version
0.221.0
[ec2-user@ip-172-31-15-198 ~]$
```

- Create a cluster named **my-ekscluster1** with one node in the same region as the EC2 machine.

```
2. 13.49.78.95 (ec2-user)
[ec2-user@ip-172-31-15-198 ~]$ eksctl create cluster \
--name my-ekscluster1 \
--region eu-north-1 \
--nodegroup-name my-nodes \
--nodes 1 \
--node-type t3.small \
--managed
2026-02-03 08:23:36 [ℹ]  eksctl version 0.221.0
2026-02-03 08:23:36 [ℹ]  using region eu-north-1
```

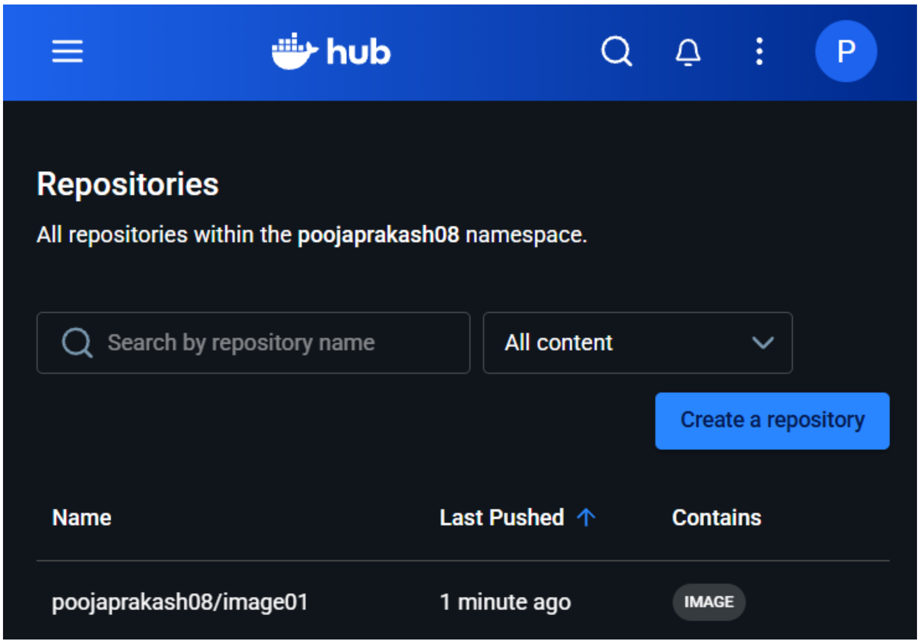
```
[ec2-user@ip-172-31-15-198 ~]$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE   VERSION
ip-192-168-5-86.eu-north-1.compute.internal   Ready    <none>   12m   v1.32.9-eks-ecaa3a6
```

- Log in to Docker Hub and push the tagged application image.

```
2. 13.49.78.95 (ec2-user)
[ec2-user@ip-172-31-15-198 ~]$ docker images
REPOSITORY   TAG      IMAGE ID       CREATED          SIZE
image01      v1       0dc11ba087e4   22 seconds ago   140MB
<none>       <none>   2d4389d8910f   41 hours ago     140MB
mysql        8.0      67fcbe35dfe8   3 days ago       786MB
[ec2-user@ip-172-31-15-198 ~]$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/

Username: poojaprakash08
Password:
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[ec2-user@ip-172-31-15-198 ~]$ docker tag image01:v1 poojaprakash08/image01:v2
[ec2-user@ip-172-31-15-198 ~]$ docker images
REPOSITORY               TAG      IMAGE ID       CREATED              SIZE
image01                  v1       0dc11ba087e4   About a minute ago   140MB
poojaprakash08/image01   v2       0dc11ba087e4   About a minute ago   140MB
<none>                   <none>   2d4389d8910f   41 hours ago         140MB
mysql                    8.0      67fcbe35dfe8   3 days ago           786MB
[ec2-user@ip-172-31-15-198 ~]$ docker push poojaprakash08/image01:v2
The push refers to repository [docker.io/poojaprakash08/image01]
05c3ab9b6383: Pushed
9c6a45fc5ba1: Pushed
b028c344a975: Pushed
5c43b6d1c516: Pushed
06f252d15777: Mounted from library/python
072aa7e4d8cf: Mounted from library/python
5662d93ce7b0: Mounted from library/python
a8ff6f8cbdfd: Mounted from library/python
v2: digest: sha256:8da4ffeeabb526d402f303947b5044841547513e9e1813f8a8888391335bc387 size: 1990
[ec2-user@ip-172-31-15-198 ~]$
```



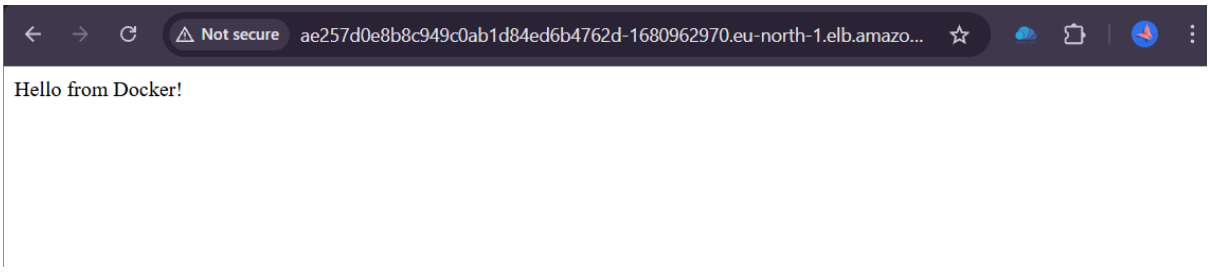
- Write a YAML file to define the Deployment and Service for the application.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      containers:
      - name: myapp-container
        image: poojaprakash08/image01:v2
        ports:
        - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: myapp-service
spec:
  type: LoadBalancer
  selector:
    app: myapp
  ports:
  - port: 80
    targetPort: 80
```

- Apply the YAML file, which creates the Deployment, Pods, and Service objects.

```
[ec2-user@ip-172-31-15-198 ~]$ vi myapp.yml
[ec2-user@ip-172-31-15-198 ~]$ kubectl apply -f myapp.yml
deployment.apps/myapp-deployment created
service/myapp-service created
[ec2-user@ip-172-31-15-198 ~]$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
myapp-deployment   2/2     2            2           15s
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myapp-deployment-64f5dfb5cb-6zjzp   1/1     Running   0          28s
myapp-deployment-64f5dfb5cb-ns6cn   1/1     Running   0          28s
[ec2-user@ip-172-31-15-198 ~]$ kubectl get services
NAME              TYPE           CLUSTER-IP       EXTERNAL-IP
               PORT(S)        AGE
kubernetes       ClusterIP      10.100.0.1       <none>
               443/TCP        65m
myapp-service    LoadBalancer   10.100.100.211   ae257d0e8b8c949c0ab1d84ed6b4762d-1680962970.eu-north-1.elb.a
mazonaws.com   80:30521/TCP   37s
[ec2-user@ip-172-31-15-198 ~]$ kubectl get svc myapp-service
NAME              TYPE           CLUSTER-IP       EXTERNAL-IP
               PORT(S)        AGE
myapp-service    LoadBalancer   10.100.100.211   ae257d0e8b8c949c0ab1d84ed6b4762d-1680962970.eu-north-1.elb.a
mazonaws.com   80:30521/TCP   87s
[ec2-user@ip-172-31-15-198 ~]$
```

- Access the application using the external IP address provided by the Service.



➢ An AWS EKS cluster was successfully created to run Kubernetes workloads. A Deployment controller was used to manage multiple replicas of the Dockerized application. The application was exposed to the internet using a LoadBalancer Service, which automatically provisioned an AWS Elastic Load Balancer.

2. L2 - Scale-up and Scale-Down the Pods Deployed

- Scale the Deployment from 2 pods to 3 pods using the `kubectl scale` command.

```
[ec2-user@ip-172-31-15-198 ~]$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
myapp-deployment   2/2     2            2           35m
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myapp-deployment-5dc95f57d5-6wkfz   1/1     Running   0          27m
myapp-deployment-5dc95f57d5-qwxq9   1/1     Running   0          27m
[ec2-user@ip-172-31-15-198 ~]$ kubectl scale deployment myapp-deployment --replicas=3
deployment.apps/myapp-deployment scaled
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myapp-deployment-5dc95f57d5-4fxtq   1/1     Running   0          14s
myapp-deployment-5dc95f57d5-6wkfz   1/1     Running   0          28m
myapp-deployment-5dc95f57d5-qwxq9   1/1     Running   0          28m
[ec2-user@ip-172-31-15-198 ~]$
```

- Scale down the Deployment, reducing the number of pods to 1

```
[ec2-user@ip-172-31-15-198 ~]$ kubectl scale deployment myapp-deployment --replicas=1
deployment.apps/myapp-deployment scaled
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS        RESTARTS   AGE
myapp-deployment-5dc95f57d5-4fxtq   1/1     Terminating   0          104s
myapp-deployment-5dc95f57d5-6wkfz   1/1     Running       0          30m
myapp-deployment-5dc95f57d5-qwxq9   1/1     Terminating   0          30m
[ec2-user@ip-172-31-15-198 ~]$
```

- Using the Kubernetes Deployment controller, the application Pods were dynamically scaled up and down with the `kubectl scale` command. Kubernetes automatically created and terminated Pods while maintaining service availability.

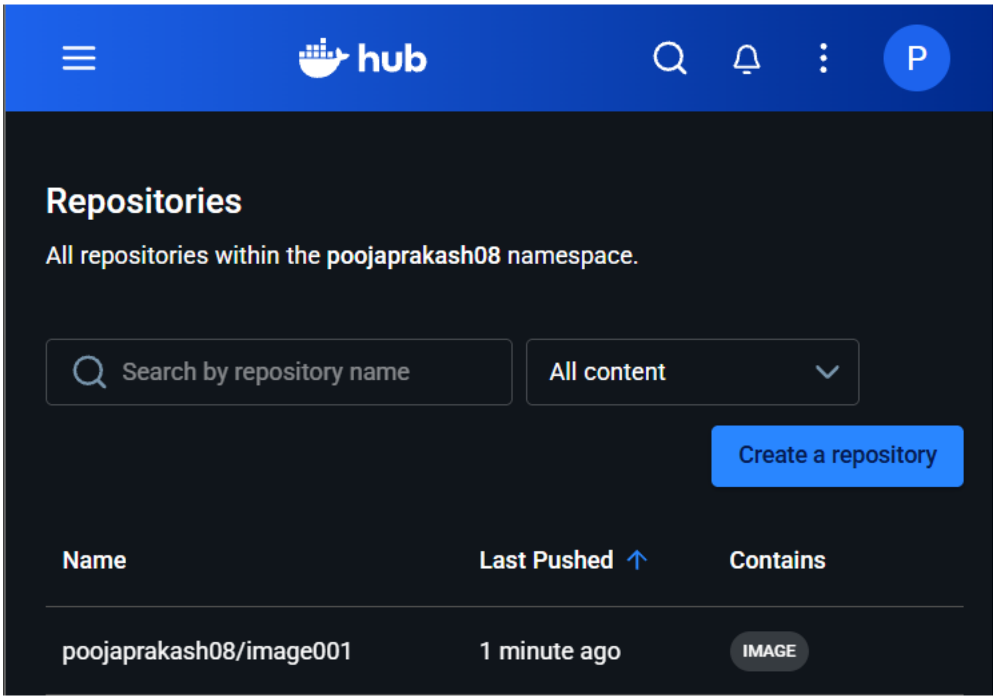
3. L3 - Implement Rolling-Update Strategy to Upgrade the Application Image from V1.0 to V1.1

- Modify the `app.py` file to introduce visible changes for the rolling update.
- Build a new Docker image with the updated files.

```
[root@ip-172-31-15-198 Docker_Assignment]# ls
Dockerfile  app.py  docker-compose.yml  requirements.txt
[root@ip-172-31-15-198 Docker_Assignment]# vi app.py
[root@ip-172-31-15-198 Docker_Assignment]# docker build -t image002:v1 .
[+] Building 0.6s (10/10) FINISHED                                                              docker:default
 => [internal] load build definition from Dockerfile                                                      0.0s
 => => transferring dockerfile: 259B                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                       0.4s
 => [internal] load .dockerignore                                                                         0.0s
 => => transferring context: 2B                                                                           0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:d0d43a8b0c352c215cd1381f3f4d7ac34cf3440cd041     0.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:d0d43a8b0c352c215cd1381f3f4d7ac34cf3440cd041     0.0s
 => [internal] load build context                                                                         0.0s
 => => transferring context: 558B                                                                         0.0s
 => CACHED [2/5] WORKDIR /app                                                                             0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                  0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                       0.0s
 => [5/5] COPY app.py .                                                                                   0.0s
 => exporting to image                                                                                    0.0s
 => => exporting layers                                                                                   0.0s
 => => writing image sha256:0e71214660884bf7c6e4ada5f72466ee38d46ce5274211793419baa9391053f3              0.0s
 => => naming to docker.io/library/image002:v1                                                            0.0s
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY                TAG       IMAGE ID       CREATED         SIZE
image002                  v1        0e7121466088   5 seconds ago   140MB
image01                   v1        0dc11ba087e4   2 hours ago     140MB
poojaprakash08/image01    v2        0dc11ba087e4   2 hours ago     140MB
<none>                    <none>    2d4389d8910f   42 hours ago    140MB
mysql                     8.0       67fcbe35dfe8   3 days ago      786MB
[root@ip-172-31-15-198 Docker_Assignment]#
```

- Tag the new Docker image and push it to Docker Hub.

```
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY                 TAG       IMAGE ID       CREATED          SIZE
image002                   v1        0e7121466088   5 seconds ago    140MB
image01                    v1        0dc11ba087e4   2 hours ago      140MB
poojaprakash08/image01     v2        0dc11ba087e4   2 hours ago      140MB
<none>                     <none>    2d4389d8910f   42 hours ago     140MB
mysql                      8.0       67fcbe35dfe8   3 days ago       786MB
[root@ip-172-31-15-198 Docker_Assignment]# docker tag image002:v1 poojaprakash08/image001:v1.1
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY                 TAG       IMAGE ID       CREATED              SIZE
image002                   v1        0e7121466088   About a minute ago   140MB
poojaprakash08/image001    v1.1      0e7121466088   About a minute ago   140MB
image01                    v1        0dc11ba087e4   2 hours ago          140MB
poojaprakash08/image01     v2        0dc11ba087e4   2 hours ago          140MB
<none>                     <none>    2d4389d8910f   42 hours ago         140MB
mysql                      8.0       67fcbe35dfe8   3 days ago           786MB
[root@ip-172-31-15-198 Docker_Assignment]# docker push poojaprakash08/image001:v1.1
The push refers to repository [docker.io/poojaprakash08/image001]
66a0c49b144e: Pushed
9c6a45fc5ba1: Mounted from poojaprakash08/image01
b028c344a975: Mounted from poojaprakash08/image01
5c43b6d1c516: Mounted from poojaprakash08/image01
06f252d15777: Mounted from poojaprakash08/image01
072aa7e4d8cf: Mounted from poojaprakash08/image01
5662d93ce7b0: Mounted from poojaprakash08/image01
a8ff6f8cbdfd: Mounted from poojaprakash08/image01
v1.1: digest: sha256:4a760ce535313fb667810fd56209dd5d4169ef5e0091b8fa9a2ac3cd70b47b84 size: 1990
[root@ip-172-31-15-198 Docker_Assignment]#
```

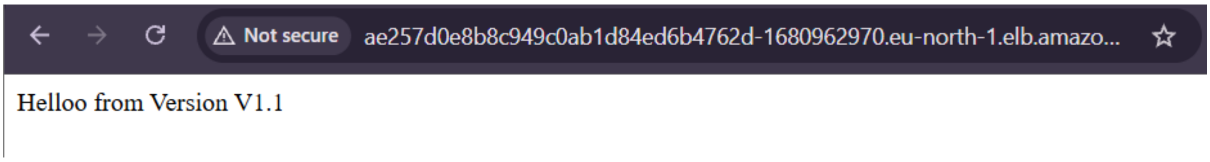


- Update the Deployment YAML file to reference the new image version.

```
      app: myapp
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: myapp
    spec:
      containers:
      - image: poojaprakash08/image001:v1.1
        imagePullPolicy: IfNotPresent
        name: myapp-container
        ports:
        - containerPort: 5000
          protocol: TCP
```

```
[ec2-user@ip-172-31-15-198 ~]$ kubectl edit deployment myapp-deployment
deployment.apps/myapp-deployment edited
[ec2-user@ip-172-31-15-198 ~]$ kubectl describe deployment myapp-deployment
Name:                   myapp-deployment
Namespace:              default
CreationTimestamp:      Tue, 03 Feb 2026 09:34:04 +0000
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 3
Selector:               app=myapp
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  0 max unavailable, 1 max surge
Pod Template:
  Labels:  app=myapp
  Containers:
   myapp-container:
    Image:        poojaprakash08/image001:v1.1
    Port:         5000/TCP
    Host Port:    0/TCP
```

- Access the application again using the same LoadBalancer URL to verify the changes.



➢ A rolling update strategy was implemented using a Kubernetes Deployment to upgrade the application image from version v1.0 to v1.1. Kubernetes gradually replaced old Pods with new Pods while ensuring zero downtime. The update and rollback process was managed using `kubectl rollout` commands.